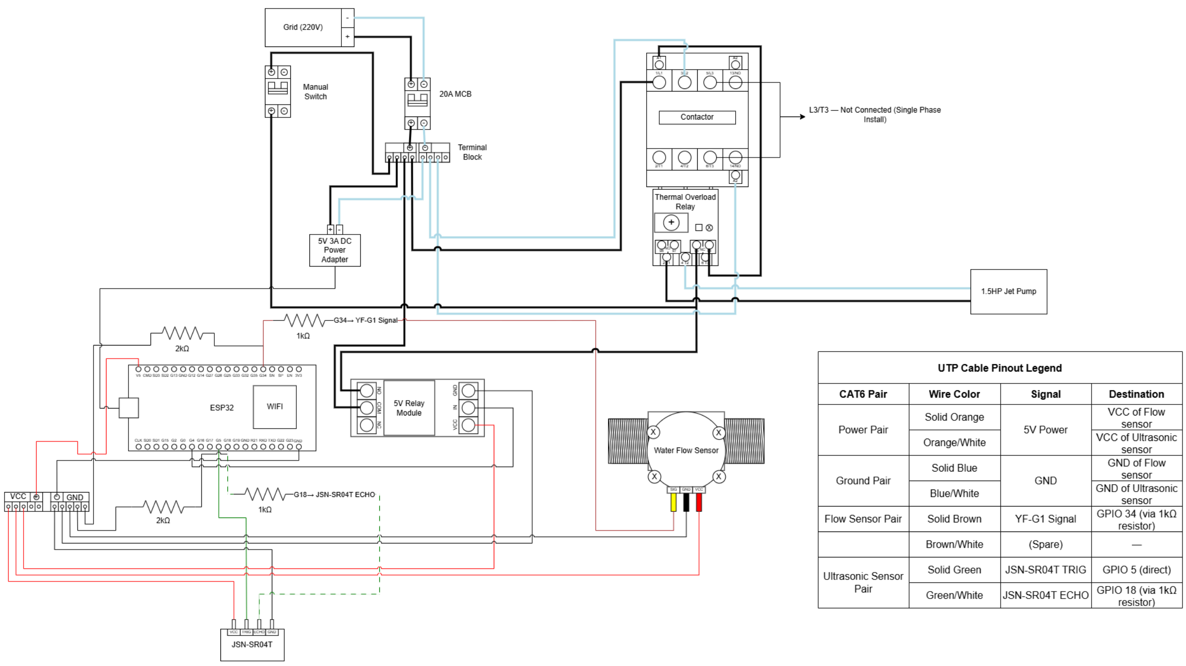
Hardware Documentation: Smart Water Pump Controller System

1. Bill of Materials (BoM) & Specifications

Item	Component	Specification / Rating	System Role
1	ESP32 Development Board	38-Pin, 3.3V Logic, Wi-Fi	The main logic controller (The Brain).
2	1-Channel Relay Module	5V DC, 10A AC Rating, Opto-isolated	Bridges low-voltage logic to the high-voltage trigger.
3	YF-G1 Water Flow Sensor	1-inch, 5V DC	Detects dry-run conditions in the pipe.
4	Waterproof Ultrasonic Sensor with Separate Probe	Waterproof, 5V DC	Measures the water level in the 600L tank.
5	AC Magnetic Contactor	CJX2-2510, 25A Rated Operational Current, 40A Conventional Thermal Current, 220V Coil	The heavy-duty switch for the 1.5HP motor.
6	Miniature Circuit Breaker (MCB)	20A, 2-Pole, 220V	Short-circuit and overcurrent protection.
7	IP65 Enclosure	30x40x20 cm, ABS Plastic	Waterproof housing for the control system.
8	DC Power Adapter	5V, 3A	Supplies stable power to the ESP32 and sensors.
9	DIN Rail	35mm wide, cut to fit	Mounting track for the MCB and Contactor.

10	Thermal Overload Relay (TOR)	LR2-D13 7A - 10A Adjustable	Protects the motor from overheating/jamming.
11	Cable Glands	PG9 (Sensors) & PG16 (Power)	Seals cable entry points against water/insects.
12	Hardware Shop Additions	10 or 12 AWG THHN Copper Wire	Required for all 220V interior connections.
13	Passive Components	1kΩ & 2kΩ Resistors	Drops the 5V sensor signal to 3.3V for the ESP32.
14	CAT6 UTP Cable	CAT6 Outdoor Gigabit Ethernet (40M)	Connection for the sensors to from the tank to the ESP32

2. System Architecture



UTP Cable Pinout Legend

CAT6 Pair	Wire Color	Signal	Destination
Power Pair	Solid Orange	5V Power	VCC of Flow sensor
	Orange/White		VCC of Ultrasonic sensor
Ground Pair	Solid Blue	GND	GND of Flow sensor
	Blue/White		GND of Ultrasonic sensor
Flow Sensor Pair	Solid Brown	YF-G1 Signal	GPIO 34 (via 1kΩ resistor)
	Brown/White	(Spare)	—
Ultrasonic Sensor Pair	Solid Green	JSN-SR04T TRIG	GPIO 5 (direct)
	Green/White	JSN-SR04T ECHO	GPIO 18 (via 1kΩ resistor)

3. Assembly Instructions

CRITICAL SAFETY WARNING: Ensure all grid power is completely disconnected before beginning the assembly. Verify with a multimeter.

Phase 1: Mechanical Assembly (The Enclosure)

1. **Prepare the Box:** Drill holes in the bottom of the 30x40x20cm IP65 enclosure. Install the PG16 glands for the thick 220V input/output wires, and the PG9 glands for the UTP LAN sensor cable.
2. **Mount the DIN Rail:** Cut the DIN rail to length and screw it horizontally across the top-middle of the enclosure's backplate.
3. **Assemble the Motor Starter:** Take the TOR and slide its three thick upper copper prongs firmly into the bottom T1, T2, and T3 terminals of the Contactor. Tighten the screws securely.

   NOTE: The LR2-D13 is a 3-phase device; for this single-phase installation, only two poles (T1 and T2) are used. Consult the LR2-D13 datasheet to confirm correct single-phase wiring and that current sensing is configured for two-wire use.
4. **Mount High-Voltage Components:** Snap the 20A MCB and the assembled Contactor+TOR onto the DIN rail side-by-side.
5. **Mount Low-Voltage Components:** In the bottom section of the box, use standoffs or mounting tape to secure the ESP32, 5V Relay Module, and the 5V Power Adapter.

Phase 2: High-Voltage Power Routing

1. **Main Input:** Route the 220V Grid Live and Neutral wires through the first PG16 gland into the TOP terminals of the 20A MCB.
2. **MCB to Contactor:** Connect short wires from the BOTTOM terminals of the MCB to the TOP L1 and L2 terminals of the Contactor.
3. **Powering the Brain:** Install a small DIN-mount terminal block (e.g., WAGO 221 series or equivalent screw terminal strip) connected to the BOTTOM terminals of the MCB. Run separate wires from the terminal block to both the Contactor (L1/L2) and the 5V Power Adapter input. Do NOT double-wire directly into MCB terminals, as this risks loose connections and arcing at 220V.

   ALTERNATIVE: Splice the 220V input prongs/wires of your 5V 3A Power Adapter directly into the BOTTOM terminals of the MCB (sharing the connection with the Contactor wires).

4. **Output to Pump:** Route your 50m drop wire through the second PG16 gland. Connect the pump's Live wire to the T1 terminal on the bottom of the TOR, and the Neutral wire to the T2 terminal on the TOR.
5. **Grounding:** Connect a green/yellow wire from the pump casing to the metal DIN rail, and then route it back to your house's main Earth Ground terminal.

Phase 3: The "Failsafe" Trigger Wiring

1. **Power the Logic:** Connect the 5V output (+) and (-) from the Power Adapter to the ESP32 (VIN and GND) and the Relay Module (VCC and GND).
2. **ESP32 to Relay:** Connect ESP32 GPIO 4 to the Relay Module IN pin.
3. **The Kill-Switch Route:**
    - Run a wire from the MCB Bottom Live terminal to the 5V Relay COM (Common) terminal.
    - Run a wire from the 5V Relay NO (Normally Open) terminal to the NC 95 pin on the TOR.
    - Run a wire from the NC 96 pin on the TOR to the Contactor A1 (Coil) terminal.
    - Run a wire from the MCB Bottom Neutral terminal to the Contactor A2 (Coil) terminal.
    - **Manual Bypass:** Wire your existing manual switch in parallel with the relay NO–COM path. Connect one terminal of the switch to the MCB Bottom Live, and the other terminal to the TOR NC 95 pin. With this wiring, flipping the switch to ON will energize the contactor coil regardless of ESP32 or relay state, providing full manual pump operation as a failsafe.

      NOTE:

      When operating in manual bypass mode, all automatic protections via the ESP32 — dry-run detection, tank level cutoff — are bypassed. The TOR thermal protection remains active at all times.

If the motor overheats, the TOR physically breaks the connection between 95 and 96, instantly killing power to the Contactor coil.

Phase 4: Sensor Integration (The 40m UTP Run)

CAT6 Pair	Wire Color	Signal	Destination
Power Pair	Solid Orange	+5V Power	VCC of Flow and Ultrasonic sensors
	Orange/White		
Ground Pair	Solid Blue	GND	GND of Flow and Ultrasonic sensors
	Blue/White		
Flow Sensor Pair	Solid Brown	YF-G1 Signal	Goes to GPIO 34 (via 1kΩ resistor)
	Brown/White	(Spare)	—
Ultrasonic Sensor Pair	Solid Green	JSN-SR04T TRIG	Goes to GPIO 5 (direct)
	Green/White	JSN-SR04T ECHO	Goes to GPIO 18 (via 1kΩ resistor)

1. **Powering Sensors:** Send 5V and Ground from the ESP32 to the water tank using the UTP wire pairs (e.g., Solid Orange for 5V, Orange/White for GND). Connect this power to both the YF-G1 and the JSN-SR04T.

2. **Flow Sensor Signal:** Connect the yellow signal wire of the YF-G1 to a UTP wire (e.g., Solid Brown), and route it to ESP32 GPIO 34.

   CRITICAL: The YF-G1 outputs a 5V signal. Do NOT connect this wire directly to GPIO 34. Inside the box, build a voltage divider: solder a 1kΩ resistor to the end of the UTP wire, connect the other end to ESP32 GPIO 34, then solder a 2kΩ resistor from GPIO 34 to GND. This drops the 5V pulse to a safe 3.3V.

3. **Ultrasonic TRIG:** Connect the TRIG pin to a UTP wire (e.g., Solid Green) and route it to ESP32 GPIO 5.

4. **Ultrasonic ECHO (Voltage Divider Fix):**
   - Connect the ECHO pin to a UTP wire (e.g., Green/White) leading back to the box.
   - Inside the box: Do not connect this wire directly to the ESP32.
   - Solder a 1kΩ resistor to the end of the UTP wire. Connect the other end of that resistor to ESP32 GPIO 18.
   - Solder a 2kΩ resistor between GPIO 18 and GND. This drops the 5V echo pulse to a safe 3.3V.

4. IPre-Energization Checklist

1. **Tug Test:** Aggressively pull on every 220V wire connection. If any wire slips, re-strip and tighten it.
2. **TOR Calibration:** Check the motor nameplate for its Full Load Amperage (FLA). Turn the dial on the front of the Thermal Overload Relay to match the motor's rated FLA (typically 8–9A for a 1.5HP 220V motor). Setting it below FLA will cause nuisance trips under normal load.
3. **Continuity Check:** Use a multimeter to ensure there is no short circuit between the Live and Neutral terminals on the MCB.